Emmanuel Flores

Ph.D. Candidate in Soft Matter Physics

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Education

2023–Present **Ph.D. in Physics**, *Tufts University*, Medford, MA, USA
Expected completion: 2028. Advisor: Prof. Timothy J. Atherton.

2023 **M.S. in Physics**, *Universidad Autónoma Metropolitana, Iztapalapa*, Mexico City

2020 **B.S. in Physics**, *Universidad Autónoma Metropolitana, Iztapalapa*, Mexico City

Academic & Research Appointments

2023–Present **Graduate Research & Teaching Assistant**, *Tufts University*

- Research (Atherton Group): develop Morpho simulations of negative tactoids, ferroelectric nematic materials and bee swarms based on liquid crystal theory.
- Teaching: TA for PHY 0011 and PHY 0012, calculus-based introductory mechanics and electromagnetism.
- Full-time Research Assistant in 2023–2024; since then, alternating between Research and Teaching Assistant appointments.

Spring 2025–Present **Seminar Organizer**, *Dept. of Physics & Astronomy, Tufts University*

- Organize the Condensed Matter Seminar series and facilitate academic discussion and guest presentations.

Spring 2026 **Graduate Instructor**, *Dept. of Physics & Astronomy, Tufts University*

- Sole instructor for the departmental Journal Club (PHY 0107), with about 20 graduate students enrolled.

Languages

Spanish (native), English (fluent; full professional proficiency)

Technical Skills

	Level	Skill	Comment
Programming:	■■■■□	Morpho	*Negative tactoid & ferroelectric nematic models*
	■■■□□	Python	*FEniCS & HOOMD-blue simulations*
	■■■□□	Mathematica	*Symbolic work & numerical PDEs*
Familiar	Julia, C, Go, Rust, Swift		
Methods	Finite-element analysis, high-performance computing, mathematical modeling		
Research	Soft matter, liquid crystals, ferroelectric nematics, optimization theory		

Conferences, Workshops & Presentations

May 2026 **Poster**, *Smectics and Distorted Nematics: Analytics, Numerics, and Experiments*, Les Houches, France
- "Shaping the Swarm: A Q-tensor Approach to the Simulation of Bee Swarms"

July 5–11, 2025 **Talk & Poster**, *Liquid Crystals Gordon Research Seminar & Conference*, NH, USA
- "Microseparation and Tactoid Formation in Liquid Crystal Nanocomposites", presented as a talk at the Seminar and a poster at the Conference.

March 2025 **Talk**, *APS Global Physics Summit*, Anaheim, CA
- "Ferronematics in Confined Geometries"

June 2–6, 2024 **Participant**, *Summer School on Soft Solids and Complex Fluids*, Amherst, MA

Publications

[1] Tom Shneer, **J Emmanuel Flores-Calderón**, Jocelyn Ochoa, Alauna C Wheeler, Isabella C Reyes, Chaitanya Joshi, Benjamin J Stokes, Linda S Hirst, and Timothy J Atherton. Catching the wave: particle transport by a moving phase boundary. *Soft Matter*, 22(2):546–554, 2026. doi:10.1039/d5sm00821b.

[2] Samia I Liba, DaYeon Lee, **J Emmanuel Flores-Calderón**, Anca S Andrei, James H Adler, Timothy J Atherton, and Linda S Hirst. Formation of tactoidal nanoparticle condensates via an aqueous liquid crystal. *npj Soft Matter*, 2(1):20, 2026.

[3] Jocelyn Ochoa, Alauna C Wheeler, Nicholas Javier C Licauco, Jasmine Q Vu, **J Emmanuel Flores-Calderón**, Timothy J Atherton, Benjamin J Stokes, and Linda S Hirst. Combining fluorescence microscopy and machine learning for a powerful method to analyse nematic domain growth in quantum dot nanocomposites. *Liquid Crystals*, 53(1):22–33, 2026. doi:10.1080/02678292.2025.2602705.

[4] F Gael Segura-Fernández, Erick F Serrato-García, **J Emmanuel Flores-Calderón**, and Orlando Guzmán. Dynamics of nanoparticle self-assembly by liquid crystal sorting in two dimensions. *Frontiers in Physics*, 9:636288, 2021.

[5] **J Emmanuel Flores-Calderón** and Timothy J Atherton. A local saddle-point theory for finite-element modeling of ferroelectric nematics. In preparation.